



OBJECT ORIENTED SOFTWARE ENGINEERING

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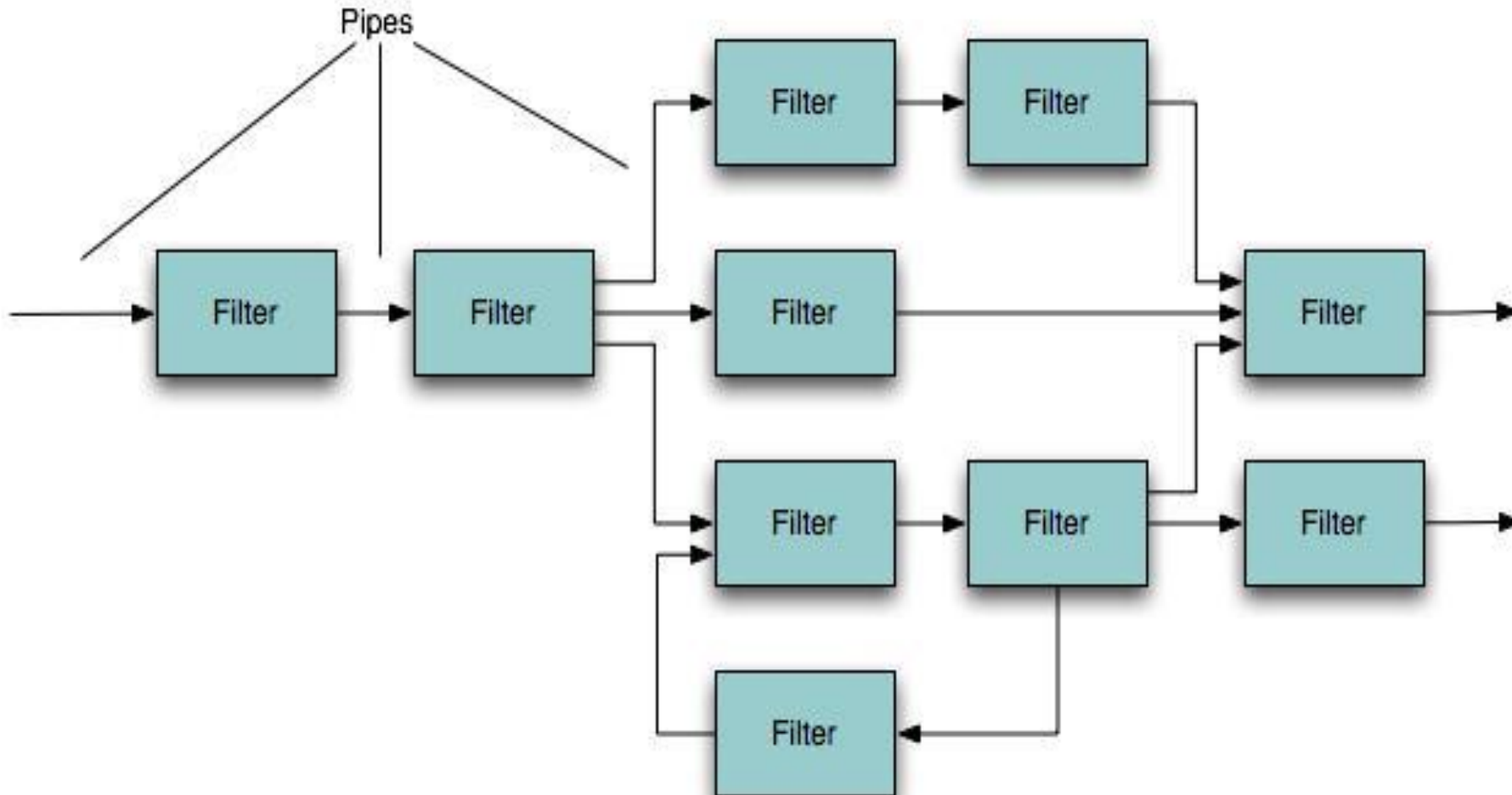
Architectural Styles

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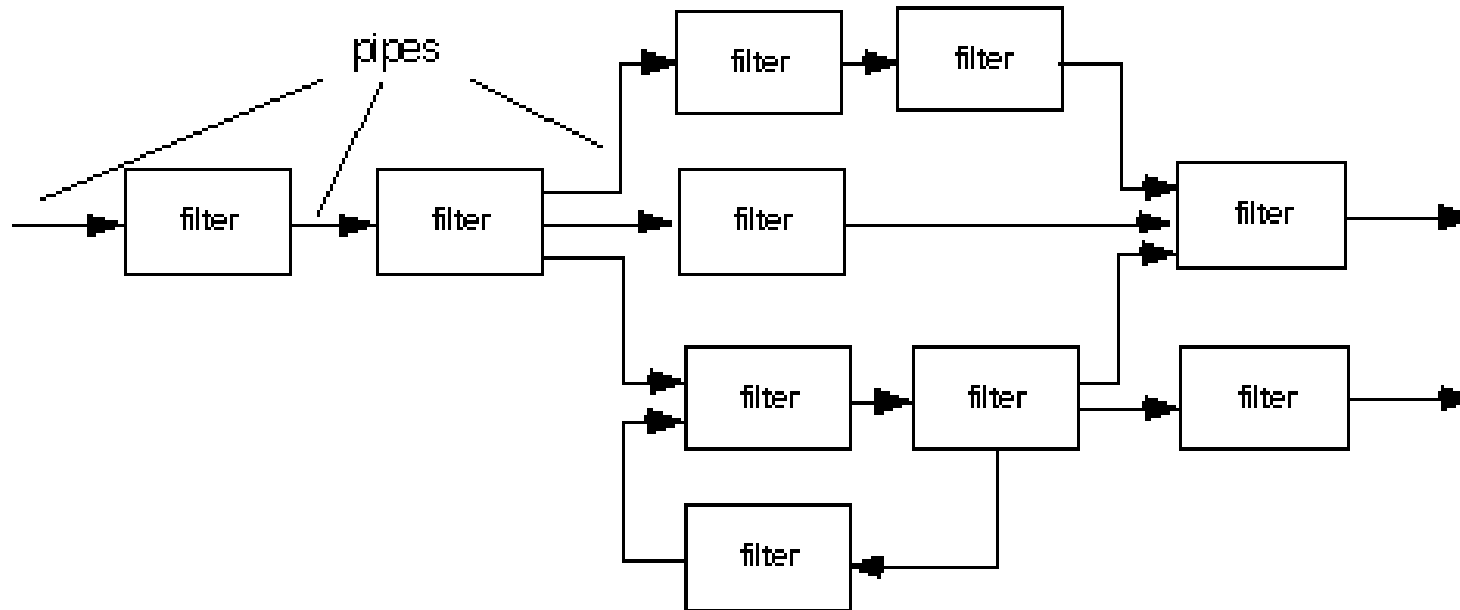
1. **Data-centered architectures**
2. **Data flow architectures**
3. **Call and return architectures**
4. **Object-oriented architectures**
5. **Layered architectures**

- Has the goal of **modifiability**
- Characterized by **viewing the system** as a **series of transformations on successive pieces of input data**.
- **Data enters the system and then flows through the components one at a time until they are assigned to output or a data store.**
- This architecture is applied when input data are to be transformed through a series of computational or manipulative components into output.
- Two well-known approaches:
 1. **Pipe and Filter**
 2. **Batch Sequential**

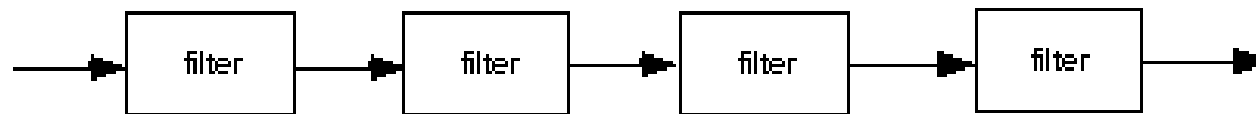


Pipe-and-filter style

- The Pipe-and-filter has a set of components, called **filters**, connected by **pipes** that transmit data from one component to the next.
- Each filter works independently of those components upstream & downstream, is designed to expect data input of a certain form and produces data output (to the next filter) of a specified form.
- The **pipes are stateless** and simply exist **to move data between filters**



(a) pipes and filters



(b) batch sequential

Batch sequential style

- If the **data flow degenerates into a single line of transforms**, it is termed **batch sequential**
- This structure **accepts a batch of data** & then **applies a series of sequential components** (filters) to transform it.
- Each step runs to completion before the next step begins

- Batch sequential compilation was regarded as a sequential process in 1970.
- In Batch sequential, separate programs are executed in order and the data is passed as an aggregate from one program to the next.
- It is a classical data processing model.

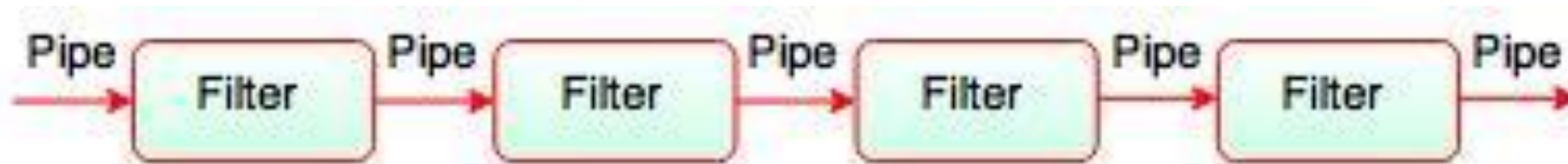


Fig. Batch Sequential



THANK YOU

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